

D1A

$$\lim_{x \rightarrow 0^+} \frac{-x^2 \sin(x-x^2)}{2(1+x)^{2/x} \cdot \ln(1-3x^3)} = \frac{-x^2 \sin(x-x^2)}{2(1+x)^{2/x} \cdot \frac{\ln(1-3x^3)}{-3x^3} \cdot -3x^3} = \frac{-x^2 \sin(x-x^2)}{2(1+x)^{2/x} \cdot -3x^3} = \frac{x^2 \sin(x-x^2)}{2e^2 \cdot -3x^3}$$

$$\frac{x^2 \sin(x-x^2)}{2e^2 \cdot -3x^3} = \frac{x^2 \sin(x-x^2)}{-6e^2 x^3} = \frac{x^2 \cdot \sin(x-x^2) \cdot x-x^2}{-6e^2 x^3} = \frac{x^2(x-x^2)}{-6e^2 x^3} = \frac{x^3 - x^4}{-6e^2 x^3} = \frac{x^3(1-x)}{-6e^2 x^3}$$

$$\frac{x^3(1-x)}{-6e^2 x^3} = \frac{1-x}{-6e^2} = \frac{1}{-6e^2}$$

D1B

$$\lim_{x \rightarrow 0^+} \frac{-2x(1-x)^{1/x} \ln(1+2x-x^2)}{3(1-\cos 3x)} = \frac{-2x \cdot \frac{1}{e} \cdot \frac{\ln(1+2x-x^2)}{2x-x^2} \cdot 2x-x^2}{3 \cdot \left(\frac{1}{2} \cdot 3x\right)} = \frac{-2x \cdot \frac{1}{e} \cdot (2x-x^2)}{\frac{9}{2}x}$$

$$\frac{-2x \cdot \frac{1}{e} \cdot (2x-x^2)}{\frac{9}{2}x} = \frac{-2x \cdot \left(\frac{2x}{e} - \frac{x^2}{e}\right)}{\frac{9}{2}x} = \frac{-4x^2 + \frac{2x^3}{e}}{\frac{9}{2}x^2} = \frac{-4x^2 + 2x^3}{x} \cdot \frac{2}{9x^2} = \frac{-8x^2 + 4x^3}{9x^2}$$

$$\frac{-8x^2 + 4x^3}{9x^2} = \frac{4x(-2x+x^2)}{9x^2} = \frac{4(-2x+x^2)}{9x} = \frac{-8x+4x^2}{9x} = \frac{4x(-2+x^2)}{9x} = \frac{-8+4x^2}{9} = \frac{-8}{9}$$

Die

$$\lim_{x \rightarrow 0^+} \frac{(1+x)^{1/3} \ln(1-3x^2)}{-x \sin(4x-3x^2)} = \frac{e^{1/3} \cdot \frac{\ln(1-3x^2)}{-3x^2} \cdot -3x^2}{-x \frac{\sin(4x-3x^2)}{4x-3x^2} \cdot 4x-3x^2} = \frac{-3x^2 e^{1/3}}{-x(4x-3x^2)} = \frac{-3x^2 e^{1/3}}{-4x^2+3x^3} = \frac{-3x^2 e^{1/3}}{-x^2(-4+3x)} = \frac{-3 e^{1/3}}{-4+3x} = \frac{-3 e^{1/3}}{-4} = \frac{3 e^{1/3}}{4}$$

2A

$$\sum_{n=1}^{+\infty} 2 \cdot \left(\frac{1}{3}\right)^n = 2 \cdot \frac{1}{3} + 2 \cdot \frac{1}{9} + 2 \cdot \frac{1}{27} + 2 \cdot \frac{1}{81} = \frac{54+18+6+1}{81} = \frac{60+19}{81} = \frac{79}{81}$$

2B

$$\sum_{n=1}^{+\infty} 3 \cdot \left(\frac{1}{4}\right)^n = \frac{3}{4} + \frac{3}{16} + \frac{3}{64} = \frac{48+12+3}{64} = \frac{53}{64}$$

3A

$$en = n! - 3^n \cdot 2 \cdot n!$$

$$ln = n^n - 6^n + \sin n \cdot 2 \cdot n^n$$

$$\frac{en}{ln} = \frac{n!}{n^n} = 0 \quad \text{uf}(n) \ll n^e \ll e^n \ll n! \ll n^n$$

$$en = o(ln)$$

D3B

$$en = n^2 - 7^n + \sin(n) \sim n^2$$

$$ln = n! - 9^n \sim n!$$

$$\frac{en}{ln} = \frac{n^2}{n!} = +\infty$$

$$\frac{ln}{en} = \frac{n!}{n^2} = 0$$

$$ln = o(en)$$

D6A

$$y = \frac{x}{x^2 + x + 9}$$

$$y' = \frac{(1)(x^2 + x + 9) - (x)(2x + 1)}{(x^2 + x + 9)^2} = \frac{x^2 + x + 9 - (2x^2 + x)}{(x^2 + x + 9)^2} = \frac{\cancel{x^2} + \cancel{x} + 9 - 2\cancel{x^2} - \cancel{x}}{(x^2 + x + 9)^2} = \frac{-x^2 + 9}{(x^2 + x + 9)^2}$$

$$\frac{-x^2 + 9}{(x^2 + x + 9)^2} \geq 0$$

$$\begin{cases} -x^2 + 9 \geq 0 \\ x^2 + x + 9 > 0 \end{cases} \quad \begin{cases} -x^2 \geq -9 \\ \forall x \end{cases} \quad \begin{cases} x^2 \leq 9 \\ \forall x \end{cases} \quad \begin{cases} -3 \leq x \leq 3 \\ \forall x \end{cases}$$

D 6 D

$$y = \frac{x}{x^2 + x + 16}$$

$$y' = \frac{x^2 + x + 16 - (x)(2x + 1)}{(x^2 + x + 16)^2} = \frac{x^2 + x + 16 - 2x^2 - x}{(x^2 + x + 16)^2} = \frac{-x^2 + 16}{x^2 + x + 16}$$

$$\frac{-x^2 + 16}{x^2 + x + 16} \leq 0$$

$$-x^2 + 16 \leq 0 \Rightarrow x^2 \geq 16 \Rightarrow x \leq -4 \vee x \geq 4$$

D 6 e

$$y = \frac{x^2 - 3}{x - 2}$$

$$y' = \frac{(2x)(x - 2) - (x^2 - 3)}{(x - 2)^2} = \frac{2x^2 - 4x - x^2 + 3}{(x - 2)^2} = \frac{x^2 - 4x + 3}{(x - 2)^2}$$

$$\frac{x^2 - 4x + 3}{(x - 2)^2} \geq 0$$

$$x^2 - 4x + 3 \geq 0 \Rightarrow x \leq -1 \vee x \geq 3$$

D 7 A

$$\int_0^1 (1 + e^{2x}) = \int_0^1 1 + \int_0^1 e^{2x} = [x]_0^1 + \frac{1}{2} \int_0^1 e^y = [1 - 0] + \frac{1}{2} [e^{2x}]_0^1 = 1 + \frac{1}{2} (e^2 - e^0) = 1 + \frac{e^2}{2} - \frac{1}{2} = \frac{e^2 - 1 + 2}{2} = \frac{e^2 + 1}{2}$$

$y = 2x$
 $dy = 2 dx$
 $\frac{dy}{2} = dx$

D 7 B

$$\int_1^2 \left(x - \frac{1}{x} \right) = \int_1^2 x - \int_1^2 \frac{1}{x} = \left[\frac{x^2}{2} \right]_1^2 - \left[\ln x \right]_1^2 = \left[2 - \frac{1}{2} \right] - (\ln(2) - \ln(1)) = \frac{3}{2} - \ln 2$$

D 7 C

$$\int_0^2 x + e^{-x} = \int_0^2 x + \int_0^2 e^{-x} = \left[\frac{x^2}{2} \right]_0^2 + \left(-\int e^y \right) = 2 + \left[-\frac{1}{e^x} \right]_0^2 = 2 + \left[-\frac{1}{e^2} + 1 \right] = 3 - \frac{1}{e^2}$$

$$\begin{aligned} dy &= -dx \\ -dy &= +dx \end{aligned}$$

Eg 1A

$$y = \frac{x+5}{x^2-9}$$

DOMINIO
 $x^2 - 9 \neq 0$
 $x \neq \pm 3$

limiti

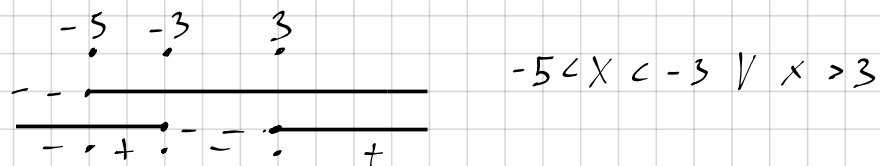
$$\lim_{x \rightarrow -\infty} \frac{x+5}{x^2-9} = 0 \quad \text{gerarchia infiniti}$$

$$\lim_{x \rightarrow +\infty} \frac{x+5}{x^2-9} = 0 \quad \text{gerarchia infiniti}$$

asintoti verticali e -3 e $+3$

Segno

$$\frac{x+5}{x^2-9} \geq 0 \Rightarrow \begin{cases} x+5 \geq 0 \\ x^2-9 \geq 0 \end{cases} \begin{cases} x \geq -5 \\ x \leq -3 \vee x \geq 3 \end{cases}$$



Monotonie

$$y = \frac{x+5}{x^2-9}$$

$$y' = \frac{(x^2-9) - (x+5)(2x)}{(x^2-9)^2} = \frac{x^2-9-(2x^2+10x)}{(x^2-9)^2} = \frac{x^2-9-2x^2-10x}{(x^2-9)^2} = \frac{-x^2-10x-9}{(x^2-9)^2}$$

$$\frac{-x^2-10x-9}{(x^2-9)^2} \geq 0 \Rightarrow \begin{cases} -9 \leq x \leq -1 \\ x \leq -3 \vee x \geq 3 \end{cases}$$



Massimi e minimi

$$\frac{-x^2 - 10x - 9}{(x^2 - 9)^2} = 0$$

$$-x^2 - 10x - 9 = 0$$

$$x = -9 \vee x = -1$$

E
g 1B

$$y = \frac{x+5}{x^2-16}$$

DOMINIO

$$x^2 - 16 \neq 0$$

$$x^2 \neq 16$$

$$x \neq \pm 4$$

Limiti

$$\lim_{x \rightarrow -\infty} \frac{x+5}{x^2-16} = 0 \quad \text{per ora che}$$

infiniti

asintoti

$$\lim_{x \rightarrow +\infty} \frac{x+5}{x^2-16} = 0 \quad \text{per ora che}$$

infiniti

orizzontali

$$\lim_{x \rightarrow -4} \frac{x+5}{x^2-16} = \frac{-4+5}{16-16} = +\infty$$

$$\lim_{x \rightarrow +4} \frac{x+5}{x^2-16} = \frac{4+5}{0} = +\infty$$

asintoti

verticali

if no

$$\frac{x+5}{x^2-16} \geq 0 \Rightarrow \begin{cases} x+5 \geq 0 \\ x^2-16 \geq 0 \end{cases} \begin{cases} x \geq -5 \\ x^2 \geq 16 \end{cases} \begin{cases} x \geq -5 \\ x \leq -4 \vee x \geq 4 \end{cases}$$

Monotonic

$$y = \frac{x+5}{x^2-16}$$

$$y' = \frac{(x^2-16) - (x+5)(2x)}{(x^2-16)^2} = \frac{x^2-16 - (2x^2+10x)}{(x^2-16)^2} = \frac{x^2-16-2x^2-10x}{(x^2-16)^2} = \frac{-x^2-10x-16}{(x^2-16)^2}$$

$$\frac{-x^2-10x-16}{(x^2-16)^2} \geq 0 \Rightarrow \begin{cases} -x^2-10x-16 \geq 0 \\ (x^2-16)^2 \geq 0 \end{cases} \begin{cases} x^2+10x+16 \leq 0 \\ x \neq \pm 4 \end{cases} \Rightarrow$$

$$x \neq \pm 4$$

-8 minimo relativo

-2 max relativo

$$\begin{array}{c} -5 \quad -4 \quad 4 \\ \hline - \quad - \quad - \quad - \quad - \\ \hline - \quad + \quad - \quad - \quad + \\ \hline -5 \leq x \leq -4 \vee x \geq 4 \end{array}$$

$$\begin{array}{c} -8 \quad -4 \quad -2 \quad 4 \\ \hline - \quad - \quad - \quad - \quad - \\ \hline \swarrow \quad \nearrow \quad \nearrow \quad \searrow \\ -8 \leq x \leq -4 \vee -4 \leq x \leq -2 \end{array}$$

Ex 2A

$$y = (x-1) \log x$$

$$\int (x-1) \log x = \int x \log x - \log x = \int x \log x - \int \log x = \int x \log x - (x \log x - x) = \int x \log x - x \log x + x$$

$$\int x \log x = \log x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} = \frac{x^2 \log x}{2} - \frac{1}{2} \int x = \frac{x^2 \log x}{2} - \frac{x^2}{4}$$

$$\frac{x^2}{2} \log x - \frac{x^2}{4} - x \log x + x = \left(\frac{x^2}{2} - x \right) \log x - \frac{x^2}{4} + x + C$$

$$F(1) = \left(\frac{1}{2} - 1 \right) \cdot 0 - \frac{1}{4} + 1 = \frac{3}{4} + C = 0$$

$$F(x) = \left(\frac{x^2}{2} - x \right) \log x - \frac{x^2}{4} + x - \frac{3}{4}$$

Una funzione è continua in x_0 se $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

$$f(x) = \begin{cases} F(x) & x > 0 \\ x + e & x < 0 \end{cases} \text{ è continua in } x = 0$$

f è continua in x_0 se $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

Si ha che $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x + e) = 0 = f(0)$. Per la continuità deve quindi essere $e = \lim_{x \rightarrow 0^+} f(x)$ cioè

$$\lim_{x \rightarrow 0^+} \left(\frac{x^2}{2} + x \right) \log x - \frac{x^2}{4} - x - \frac{3}{4} = -\frac{3}{4} = 0$$

$$\lim_{x \rightarrow +\infty} \frac{F(x)}{x^n} = \frac{\left(\frac{x^2}{2} - x \right) \log x - \frac{x^2}{4} + x - \frac{3}{4}}{x^n} \sim \frac{x^2 \log x}{2x^n} = \begin{cases} 0 & \text{per } n \geq 3 \\ +\infty & \text{per } n = 1, 2 \end{cases}$$

3A

$$\sigma_n = \sqrt{n} \left(1 - \sqrt[4]{1 - \frac{3}{n^2}} \right) \approx \sqrt{n} \cdot \frac{3}{4n^2}$$